

SEVEN-LAYER MODEL (7LM)

Authoritative Architecture Specification

Canonical research directory topology and uniform layer information sheets

Authority

This specification states the controlling 7LM research architecture: one canonical agnostic directory topology and one uniform information sheet for each layer. The structure, layer boundaries, directory prose, machine-metadata distinction, and stated rationales are authoritative for this research program. Capability branches may remain empty when unused; they remain part of the agnostic topology so a project can instantiate them without changing the architecture.

Self-description and machine-instruction rule

Every directory carries a local Layer 0 semantic description. In tree notation, /0 denotes that directory's local 0.md without drawing the file as a separate branch. The same directory separately carries machine-readable metadata for machine instructions, constraints, permissions, routing, validation, and lifecycle behavior. /0 carries human-readable meaning; metadata carries machine-actionable instruction. Neither replaces the other. This separation keeps semantic description distinct from machine operation.

Direction rule

Scientific construction proceeds from L0 outward only as dependencies are established. External inspection begins at L7 and proceeds inward. Description does not become prescription merely because the structure can be traversed.

7LM is a dependency and fault-containment architecture

7LM exists to:

1. separate ontology/semantics from state space, formal mathematics, engineering mathematics, realized work, research judgment, memory/history, and external provenance/governance/peering/publication;
2. pin lower-layer contracts so multiple research corpora can rely on them;
3. prevent semantic drift;
4. localize errors to the layer that owns them;
5. allow repair, promotion, pinning, supersession, and shedding without collapsing the whole research program;
6. keep implementation details from silently redefining lower-layer science.

The operational debugging principle is:

Find the first layer where the invariant or contract fails.

Two valid directions

Authorized external inspection inward:

L7→L6→L5→L4→L3→L2→L1→L0

Scientific construction outward:

L0→L1→L2→L3→L4→L5→L6→L7

Architecture-wide dependency rules

Higher layers do not silently repair lower layers. If a higher-layer object requires different lower-layer meaning, the lower-layer dependency is revised explicitly rather than reinterpreted in place.

Layer ownership is local. A topic may appear at several layers, but each representation belongs at the lowest layer where that representation can be stated correctly.

Change one layer at a time. A correction at one layer does not silently rewrite another layer.

A canonical branch may remain empty when unused. Presence in the agnostic topology establishes admissible capability, not mandatory population.

Directory metadata may enforce a machine rule, but it does not create the human semantic meaning carried by /O. Conversely, /O does not substitute for machine instructions.

Operational containment and analytic expansion are different relations. A complete architecture may remain owned at one 7LM layer while questions about its origin, justification, development, or effects generate representations at other 7LM layers. The contained architecture is not remapped merely because the inquiry expands.

Structured Architecture Outline

APA-like hierarchy with author notes • current working architecture

1. Purpose and controlling method

The Seven-Layer Model (7LM) is a dependency and fault-containment architecture for separating local meaning, admissibility, exact formalization, engineering approximation, realized work, research interpretation, reconstructable memory, and external presentation. Its practical debugging rule is to find the first layer at which the required invariant, evidence relation, or handoff fails, then return the correction to that owner.

Scientific construction/prescription proceeds outward: $L0 \rightarrow L1 \rightarrow L2 \rightarrow L3 \rightarrow L4 \rightarrow L5 \rightarrow L6 \rightarrow L7$, but only as far as established dependencies justify. Interpretive inspection of an externally owned object proceeds inward: $L7 \rightarrow L6 \rightarrow L5 \rightarrow L4 \rightarrow L3 \rightarrow L2 \rightarrow L1 \rightarrow L0$.

Author note. Description does not prescribe. A path can be observed or interpreted without becoming a normative requirement. Prescription requires an established dependency that has actually been tested or ratified to the degree the next layer needs.

2. Topology and traversal

L7 is outside the USO and is physically a sibling boundary to USO/. L6 through L0 are the scientific interior. Physical siblinghood and semantic sequence are different relations: crossing $L7 \leftrightarrow L6$ is a boundary/peering event, not filesystem nesting.

```
external peer / citizen / practitioner
      |
      v
    [ L7 SURFACE / FORWARD FACE ]
===== boundary / peering separator =====
              USO

    [ L6 LIBRARY / MEMORY ]
    [ L5 RESEARCH / DISCOURSE ]

    [ L4 CONVERSION / REALIZATION BAY ]
      contained operational architecture
      may expand internally here
      example: complete OSI L1-L7 stack

    [ L3 ENGINEERING MATHEMATICS ]
      [ L2 FORMAL MATHEMATICS ]
        [ L1 STATE SPACE ]
----- local L0 boundary -----
    [ L0 SEMANTIC / FOUNDATIONAL ]
```

The canonical research tree keeps the interior physically flat: l6/, l5/, l4/, l3/, l2/, l1/, and l0/ are siblings under USO/. The arrows, groupings, and visual bands in this outline indicate traversal, dependency, or analytic view; they do not create child-directory containment.

Analytic view	Meaning
L6 / L5 upper interior	Memory/history and research/discourse are the upper scientific interior. They remain separate layer owners even when shown as one research-facing band.
L4 realization bay	A whole realized architecture may be contained here and may expand internally according to its own native structure without commandeering 7LM layer numbers.
L3 / L2 / L1 technical-possibility bundle	Engineering mathematics, exact formalization, and admissible state space can be examined together when the question requires all three dependencies.

Analytic view	Meaning
L3 / L2 nested sub-bundle	Within the broader L3-L2-L1 view, L3-L2 is a tighter engineering/formal sub-bundle when L1 is already pinned and not itself under question.
L0 local semantic boundary	L0 is deliberately separated from L1. L0 establishes the local semantic/foundational contract; L1 establishes admissible states under that contract.

Author note. These bundles are semantic and dependency views, not filesystem nesting and not mandatory traversal paths. The same scientific object may be railed through different subsets of the architecture according to the question being asked, while each representation remains owned by the layer that can state it correctly.

3. Contained architectures and analytic expansion

An architecture may be contained at one 7LM layer operationally while questions about that architecture expand across 7LM epistemically. Operational ownership answers where the architecture is realized; analytic expansion answers which 7LM owner is needed to state the particular question or representation correctly.

3.1 OSI as an operating architecture

When the question is how the OSI architecture operates, the complete OSI model remains inside 7LM L4. OSI L1 through OSI L7 are internal layers of a contained realized architecture; OSI L7 is not 7LM L7, and the OSI layer numbers do not map onto the 7LM layer numbers.

3.2 OSI as an object of inquiry

When the question changes to the origins, development, justification, standardization, institutional history, or design constraints of OSI, the inquiry may rail the OSI object across the 7LM architecture. The operating OSI stack remains an L4 object; the inquiry generates layer-owned representations of that object according to what is being asked.

7LM layer	OSI-origin inquiry may ask / represent
L7 - Surface / Forward Face	Externally owned standards and source bindings, provenance, standards-publication state, public/legal governance, and relationships to standards bodies or other peers.
L6 - Library / Memory	Reconstructable development history, versions, superseded or retired material, internal chronology, and the record needed to recover how the local inquiry developed.
L5 - Research / Discourse	Institutional debate, research interpretation, committee or scholarly discourse, pedagogical framing, competing explanations, and judgments about why the architecture took its form.
L4 - Conversion / Realization	The realized OSI architecture itself as specified, implemented, taught, tested, transmitted, or otherwise made operational.
L3 - Engineering Mathematics	Engineering constraints, tradeoffs, approximation, tolerances, uncertainty, feasibility, or other engineering conditions that shaped or test the architecture.
L2 - Formal Mathematics	Exact formal relations, operators, proofs, or formal constraints used to state the architecture where such formalization is actually warranted.
L1 - State Space	Admissible structural states, possibilities, exclusions, or coordinate/state-space descriptions when these are part of the question.
L0 - Semantic / Foundational	The locally authored semantic distinctions required to describe what is being investigated. Externally owned standards definitions remain owned by their source and are bound through L7 rather than copied into L0.

Author note. The OSI model does not become 7LM L0-L7. The question about the OSI object expands. Operational containment remains L4, while representations produced by the inquiry are owned by the 7LM layer that can state them correctly.

3.3 Nested analytic views

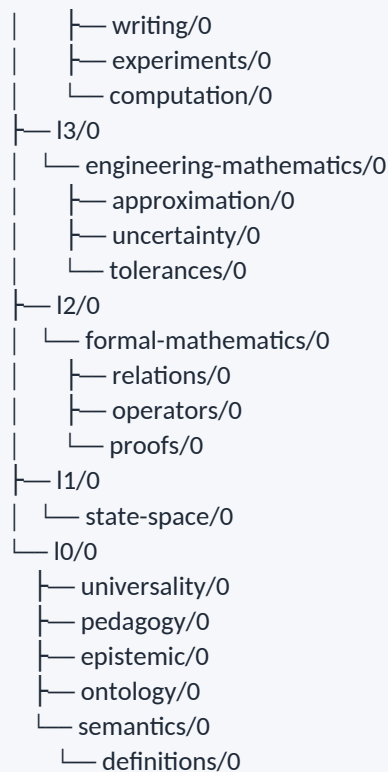
Some questions require only the L3-L2 engineering/formal sub-bundle. Others require the broader L3-L2-L1 technical-possibility bundle because admissibility itself is under examination. L0 is crossed only when the local semantic or foundational contract is itself part of the question. These are selectable analytic rails, not new containers and not a requirement that every inquiry traverse every layer.

Canonical Research Directory Topology

One complete agnostic directory structure is available to all research repositories. A branch may remain empty when unused; the canonical tree is not reduced merely because one project does not need a capability.

Notation: /0 means the directory carries its local 0.md human-readable semantic description. Directory metadata is attached to the directory itself and is therefore not drawn as another branch in the tree.

```
<repo>/
├── README.md
├── I7/0
│   ├── provenance/0
│   │   ├── prior-art/0
│   │   ├── sources/0
│   │   ├── lineage/0
│   │   └── authority/0
│   ├── governance/0
│   │   ├── copyright/0
│   │   ├── licensing/0
│   │   ├── patents/0
│   │   ├── permissions/0
│   │   ├── access/0
│   │   └── restrictions/0
│   ├── peering/0
│   │   ├── private/0
│   │   │   ├── ingress/0
│   │   │   │   ├── peer-1/0
│   │   │   │   │   └── object-1/0
│   │   │   └── egress/0
│   │   │       ├── peer-1/0
│   │   │       │   └── object-1/0
│   │   └── public/0
│   │       ├── ingress/0
│   │       │   ├── peer-1/0
│   │       │   │   └── object-1/0
│   │       └── egress/0
│   │           ├── peer-1/0
│   │           │   └── object-1/0
│   └── publications/0
├── uso/0
│   ├── I6/0
│   │   ├── history/0
│   │   └── retired/0
│   ├── I5/0
│   │   ├── research/0
│   │   │   ├── lanes/0
│   │   │   ├── review/0
│   │   │   └── findings/0
│   │   └── institution/0
│   │       ├── peers/0
│   │       └── undergraduates/0
│   ├── I4/0
│   │   └── work/0
```



Canonical-tree rule. Unused now does not mean removed. The same complete topology is available to Autonomous, Auditonomous, OMMP, classroom research, institutional research, and future research repositories.

Each /0 begins with the object directory and path: /data/location/object

Standing. Each local 0.md first identifies the directory/object and its path, then supplies that directory's semantic description. Machine instructions belong to the directory metadata, not to the semantic prose.

L7 = Desktop

USO = USO as bounded scientific object

The USO is the compartmentalized scientific interior that allows the components/representations of a bounded thing to occupy their correct layer without forcing the entire repository or corpus to share one maturity status. General repository infrastructure is not automatically a layer.

A bounded exemplar can be written or made at L4 while its semantic, state-space, formal, engineering, research, memory/history, provenance/governance, and surface representations remain separately addressable. The layer tree is the discourse structure of a thing, not a command to refile unrelated repository infrastructure.

L0 — LAYER 0

Local semantic/rail version of the bounded thing; origin contract, not a warehouse for external definitions.

Core question	What local ontological, semantic, epistemic, and universality distinctions does THIS object establish before an admissible state is declared?	
Function	Hold the object’s locally authored foundational representation and system-level pedagogy without absorbing externally owned definitions.	
Feeds	A bounded thing plus external bindings already identified at L7, oriented through L6, and interpreted where necessary at L5.	
Output	A local rail-grounded L0 contract sufficient for L1. A rail may remain sparse or gate-only when no local representation has been authored.	
Hard boundary	Locally authored L0 definitions may be exposed at L7 and peered by other systems, but externally owned definitions are never copied inward as local meaning. L0 does not own prior art, provenance, public/legal governance, state transition, operator, proof, numerical method, experiment, or research conclusion.	
Machine metadata	Machine-readable metadata may identify the local rail, owner, version, gate status, and allowed machine actions for L0 directories. It may enforce handling constraints, but it must not define or replace the semantic meaning in /0.	
Owns / represents	Permitted actions	Acceptance / boundary check
• Ontological, Semantic, Epistemic, and Universality rail boundaries for this object. • System-level traversal/attachment pedagogy as a directory/function, without making Pedagogy a fifth rail. • Local vocabulary or semantic representation actually authored by this object.	• bind local distinctions to their owning rail • state ambiguity or unresolved meaning without filling it by guess • version a changed local distinction rather than silently rewriting prior use	• Every L1 distinction can resolve to a local L0 distinction or an explicitly named unresolved dependency. • No external definition is copied inward merely to make L0 look complete. • No neighboring rail silently supplies another rail’s local meaning.

Directory structure

```
US0/l0/0
├── universality/0
├── pedagogy/0
├── epistemic/0
├── ontology/0
├── semantics/0
│   └── definitions/0
```

Boundary / traversal rule. L0/semantics contains the semantic representation authored by the bounded thing. An externally owned definition is not copied into L0 merely to populate the directory.

Directory prose and rationale

Directory / thing	Prose	Why
uso/l0/	Layer 0 container for the bounded object’s locally authored foundational and rail representation.	Keeps foundational meaning distinct from state, formalization, engineering, realized work, research judgment, provenance, and surface exchange.
universality/	Directory for the Universality rail content owned by L0.	Preserves Universality as its own rail instead of letting another rail silently supply it.
pedagogy/	Directory for system-level traversal and attachment pedagogy.	Keeps pedagogy explicit without turning it into a fifth philosophical rail.
epistemic/	Directory for the Epistemic rail content owned by L0.	Keeps epistemic distinctions inside the rail that owns them.
ontology/	Directory for the Ontological rail content owned by L0.	Keeps ontological distinctions inside the rail that owns them.
semantics/	Directory for the Semantic rail content authored by this bounded object.	Keeps local semantic meaning separate from externally owned definitions and later formal representation.
semantics/definitions/	Directory for definitions actually authored by this object under its Semantic rail.	Prevents an external definition from being copied inward and mistaken for locally owned meaning.

MOST IMPORTANT: I THINK DRIFT LIVES HERE! NOT ANYWHERE ELSE...(NEEDS A LOT OF STUDY)

Standing. This statement is retained as a research hypothesis: semantic drift is expected to originate when the local L0 meaning contract changes or is misapplied. The hypothesis remains open to testing rather than being promoted to a settled rule.

In the case of:

In the Autonomous exemplar, IEEE/IEC have already been adjudicating relevant definitions through standards. The locally authored semantic definition used by this bounded object still lives at Layer 0, while the discourse, research, and state-space objects are distilled to their own layers. AUTONOMOUS and AUDITONOMOUS are retained as pedagogical exemplars: EXEMPLAR AUTO and EXEMPLAR AUDIT.

Standing. In this exemplar, the L0 object is the locally authored semantic definition of Autonomous. IEEE/IEC source definitions and standards remain externally owned inputs bound through L7 peering; their source identity, version, authority, and prior-art relation are represented through L7 provenance. Historical internal use remains reconstructable through L6.

Author note. Current correction: L0/semantics is the semantic version we make of the thing. If we do not make one, the directory is not populated with someone else’s definition.

L1 — STATE SPACE

Admissible states and static possibility bounds.

Core question	What states and distinctions are possible or admissible under the established L0 contract?	
Function	Freeze the board: dimensions, admissible states, invariants, exclusions, equivalence/identity conditions, and static compatibility.	
Feeds	The current L0 contract for the bounded thing.	
Output	An L1 state-space contract defining what L2 may represent and operate upon.	
Hard boundary	No carrying, transition, selection, learning, calculation, propagation, or transformation. Formal operation begins at L2.	
Machine metadata	Machine-readable metadata may identify the state-space version, pinning/immutability status, admission checks, and compatibility validation. It may enforce the frozen board, but it does not create the states or their meaning.	
Owns / represents	Permitted actions	Acceptance / boundary check
- admissible and excluded states - dimensions, domains, ranges, equivalence classes, identity conditions - static invariants and compatibility constraints	- admit, exclude, distinguish, bound, freeze - test state membership and compatibility - issue a new version when the possibility landscape changes	- Every admitted state is grounded in L0. - Invariants are mutually compatible or the collision is explicit. - Changes return to L1 and create a new version.

Directory structure

```
uso/l1/0
└─ state-space/0
```

Boundary / traversal rule. L1 establishes the admissible state space. Formal operation begins at L2.

Directory prose and rationale

Directory / thing	Prose	Why
uso/l1/	Layer 1 container for the admissible state-space representation grounded in L0.	Keeps static possibility separate from the formal operations that begin at L2.
state-space/	Directory for admissible and excluded states, dimensions, invariants, identity conditions, and static compatibility.	Establishes what may exist or be represented before lawful operation is introduced.

In the case of: State-space containment and scientific claim boundaries

9.1 They cannot leave state space — unless they explicitly justify a new one

A valid object or transition must remain inside the declared state space and its admissibility rules.

If a result falls outside the declared state space, then either:

- the construction is invalid under the current model; or
- the current state space is incomplete for the phenomenon, and a new explicitly justified extension must be proposed.

The model may not silently step outside its state space and still claim continuity with the same system.

9.2 Mathematical permission is not automatic scientific or engineering permission

A mathematical construction is allowed to remain mathematically valid even when no engineering realization exists.

The construction stops at the first unresolved required bridge.

That is not rejection. It is honest placement.

Example used in session: peer-reviewed FTL/warp-drive mathematics may be legitimate mathematics while still lacking an engineering bridge. The math can stay where it is. It does not receive a free promotion into realizable propulsion.

9.3 No exotic capability gets a free explanatory advantage

FTL, unlimited power, perfect information, zero latency, infinite endurance, and similar assumptions may be studied conditionally.

They may not be used as free empirical resources unless the capability itself carries its own evidentiary and engineering burden.

The session rule became:

Build whatever the math permits. Claim only what the layer has earned.

9.4 Observable science is bounded without forbidding speculation

The architecture is not anti-speculation.

Hypotheses may be created and mathematically explored.

The discipline is that unobserved or unengineered capability cannot be silently promoted downward into the foundational or physical claim space.

State-space and claim boundaries

An object or formal operation may not silently leave the declared state space and still claim continuity with the same L1 contract. An out-of-space result either fails under the current model or motivates an explicit proposal for a revised state space.

Mathematical permission is not automatic engineering or empirical permission. A mathematically valid object may stop at L2. Engineering feasibility must be earned at L3, realized work at L4, and scientific interpretation at L5. Packaging or publicity at L7 never promotes maturity by itself.

The architecture therefore permits speculation while making maturity explicit: build whatever the mathematics legitimately permits; claim only what the completed handoffs have earned.

Author note. STATE and IMMUTABLE remain useful descriptive/mnemonic language, but the controlling layer name is State Space.

L2 — FORMAL MATHEMATICS

Exact formal objects and lawful operation over L1.

Core question	What follows exactly when lawful operations are defined over L1?	
Function	Define notation, types, relations, operators, constructors, carriers, exact models, derivations, and proof obligations.	
Feeds	The L1 state-space contract.	
Output	An exact L2 formal contract supplying L3 with behavior, invariants, formulas, and unresolved proof obligations.	
Hard boundary	No engineering tolerance, approximation quality, physical realization, measurement, or empirical truth is claimed here.	
Machine metadata	Machine-readable metadata may identify formal artifact type, dependencies, proof/test status, domain/codomain pointers, and execution restrictions. It assists machines without replacing the formal or semantic statement.	
Owns / represents	Permitted actions	Acceptance / boundary check
• formal symbols and types • relations, functions, operators, gates, constructors, carriers, transition systems • axioms, definitions, lemmas, theorems, derivations, exact formulas	• formalize, derive, prove, type-check, construct, test closure • declare domain/codomain/partiality • mark conjectural or unproved claims explicitly	• Every symbol binds to L0 meaning and L1 admissibility. • Every operation has declared domain/codomain or partiality. • Exact results are supported by proof/derivation or marked as unproved.

Directory structure

```
uso/l2/0
├─ formal-mathematics/0
│   ├─ relations/0
│   ├─ operators/0
│   └─ proofs/0
```

Boundary / traversal rule. Engineering tolerance, numerical approximation, and numerical uncertainty belong to L3, not to L2 exact formal ownership.

Directory prose and rationale

Directory / thing	Prose	Why
uso/l2/	Layer 2 container for exact formal representation and lawful operation over L1.	Separates exact mathematical structure from engineering approximation and empirical realization.
formal-mathematics/	Directory for the exact formal system: symbols, types, models, derivations, and obligations.	Keeps formal mathematics grouped under one exact owner before engineering choices are introduced.
relations/	Directory for exact relations among admitted formal objects.	Keeps relational structure distinguishable from the operators that act and the proofs that warrant claims.
operators/	Directory for exact operators, functions, gates, constructors, and other lawful formal actions.	Keeps operation explicit rather than burying it inside prose or engineering implementation.
proofs/	Directory for proofs, lemmas, derivations, and explicit unproved obligations.	Keeps warrant for exact claims inspectable and separate from the claim's notation or later numerical realization.

Author note. Current correction: "finite tolerance" is not an L2 ownership rule. Engineering tolerance and numerical uncertainty belong at L3.

L3 — ENGINEERING MATHEMATICS

Approximation, numerical representation, tolerance, uncertainty, and feasibility.

Core question	How can the exact L2 relation be made feasible under declared constraints?	
Function	Translate formal behavior into implementable numerical/engineering methods with explicit units, precision, approximation, tolerance, uncertainty, complexity, and operating regions.	
Feeds	L2 formal contract plus declared environmental, resource, and performance constraints.	
Output	An L3 engineering specification telling L4 what can be built/run and within what conditions.	
Hard boundary	L3 is not the realized L4 output and does not itself create empirical evidence.	
Machine metadata	Machine-readable metadata may identify units, precision, approximation profile, tolerance/uncertainty bounds, implementation target, and validation conditions. It carries machine constraints while /0 carries the human semantic description.	
Owns / represents	Permitted actions	Acceptance / boundary check
- algorithms and abstract data-structure choices - units, numerical representation, precision, approximation, discretization, tolerance - error/uncertainty budgets, sensitivity, stability, convergence, complexity - simulation, optimization, feasibility arguments	- approximate, discretize, optimize, estimate, bound error, choose representation - compare candidate methods to L2 invariants - state operating assumptions and valid region	- Error/uncertainty, units, precision, and tolerance are explicit. - Required L2 invariants are preserved within declared bounds. - Resource/complexity claims are analyzed or labeled estimates.

Directory structure

```
uso/l3/0
├── engineering-mathematics/0
│   ├── approximation/0
│   ├── uncertainty/0
│   └── tolerances/0
```

Boundary / traversal rule. L3 owns engineering approximation and feasibility; it does not own the realized L4 output.

Directory prose and rationale

Directory / thing	Prose	Why
uso/l3/	Layer 3 container for numerical and engineering feasibility applied to the L2 formal contract.	Separates implementable approximation and constraint handling from exact L2 mathematics and realized L4 work.
engineering-mathematics/	Directory for algorithms, numerical representation, units, precision, feasibility, stability, and complexity.	Keeps the engineering translation of exact formal behavior explicit and inspectable.
approximation/	Directory for approximations and discretizations used to make exact relations workable.	Makes loss or approximation visible instead of letting it appear as exact mathematics.
uncertainty/	Directory for uncertainty and error characterization associated with an engineering representation.	Keeps uncertainty explicit rather than silently absorbing it into a result.
tolerances/	Directory for declared tolerances and admissible engineering bounds.	Keeps acceptance bounds separate from the exact L2 relation they approximate or implement.

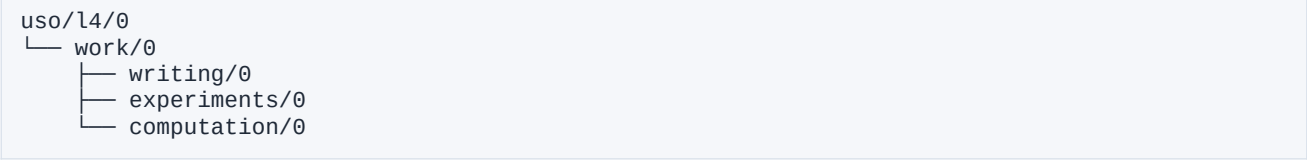
Author note. INFINITE is retained as a working term for open-ended tolerance/refinement across implementations and time, not as permission to move exact L2 objects here.

L4 — LAYER 4

Conversion / realized work / identifiable output.

Core question	What was actually made, written, run, measured, integrated, stored, transported, converted, or delivered?	
Function	Commit resources to produce an identifiable intellectual, digital, organizational, or physical work product or recorded failure.	
Feeds	Pinned L3–L0 dependency material, exact versions, obligations, demarcation, authority, and resources.	
Output	An audit-ready realized work product/evidence record with traceable inputs, transformations, conditions, tests, results, failures, and rework.	
Hard boundary	L4 does not authorize its own L5 interpretation and cannot flatten lower contracts into a convenience summary.	
Machine metadata	Machine-readable metadata may identify build/run instructions, environment, resource requirements, inputs/outputs, integrity checks, permissions, and fault routing for realized work. It does not supply the L5 interpretation of the result.	
Owns / represents	Permitted actions	Acceptance / boundary check
• writing/document production as performed work • experiments as executed, observations, measurements, data collection, tests • code execution, storage, deployment, manufacturing, integration, whole systems • resource commitments and work trace	• make, write, code, run, measure, test, collect, integrate, convert, store, transport, deploy, manufacture • verify conformance and rework against the governing input contract • record result or failure without converting it into a conclusion	• Governing lower dependencies are pinned/resolvable. • A distinguishable output or recorded failure exists. • Inputs, conditions, transformations, measurements, and demarcation are traceable. • Verification is separated from L5 judgment.

Directory structure



Boundary / traversal rule. Writing belongs to L4 as performed work. Its research meaning belongs to L5; a released publication belongs to L7. Contained architectures may expand internally without changing their 7LM owner. When the question is how the OSI architecture operates, the complete OSI L1-L7 stack remains inside 7LM L4, and OSI L7 is not 7LM L7. When the question instead concerns OSI origins, development, justification, standardization, or design constraints, the inquiry may expand across the appropriate 7LM layers without remapping the OSI stack itself. An OSI network fault is therefore an L4 operational fault unless separately shown to affect another layer’s owned object.

Directory prose and rationale

Directory / thing	Prose	Why
uso/l4/	Layer 4 container for performed and realized work that returns an identifiable output or recorded failure.	Keeps the event of making or doing distinct from the later L5 interpretation of what the result means.
work/	Directory for traceable realized work governed by pinned lower-layer dependencies.	Provides one place for the performed event without flattening the L3-L0 package that governs it. A complete internal architecture, including OSI, remains an L4 object even when it has its own internal layers.
writing/	Directory for writing and document production as performed work.	Distinguishes making the document from its L5 research meaning and its L7 released publication.
experiments/	Directory for experiments as actually executed, including observations, measurements, tests, and failures.	Distinguishes performed experiment from L3 measurement design and L5 interpretation.
computation/	Directory for computation as actually run, including executable processing and recorded outputs or failures.	Separates executed computation from the formal L2 relation and engineering L3 method that specify it. OSI application and network layers remain execution detail inside L4; OSI L7 is not the 7LM Surface / Peering layer.

MOST IMPORTANT: THE CUT LIVES HERE

Standing. The cut is the realization boundary: L3 and below specify what is admissible, exact, or engineerable; L4 is where resources are committed and an identifiable work product, measurement, execution, or failure is produced.

In the case of:

Cross-layer activity cases

Case	Placement rule
Research	L5 owns interpretation/review; L6 preserves internal memory/history; L7 owns external provenance and exposes permitted presentations; L4 performs research work events; lower layers own their specific representations.
Writing	The act/file production is L4. The research argument is L5. Internal version and correction history are L6. Public provenance, governance, and the released publication are L7.
Experiment	Executed experiment, observation, measurement, and raw work product are L4; engineering method/tolerance L3; exact relations L2; possibilities L1; interpretation/conclusion L5.
Contained architecture (OSI)	Operationally, the full OSI L1-L7 architecture remains an L4 object. Questions about its origins, development, justification, standards history, or design constraints may be railed through other 7LM layers; this analytic expansion does not remap OSI layers to 7LM layers.
Code	Algorithmic/engineering representation can be L3; formal proof/model L2; execution, deployed bytes, storage, interfaces, and operational result L4.
Data	Collected/stored instances and transformations are L4; engineering representation/precision L3; formal encoding may be L2; state categories L1; external source/lineage is L7 provenance; internal development history is L6; interpretation is L5.
External definition	Owner remains external at L7 peering; source/version/authority/prior-art relation is L7 provenance; applicable legal/access conditions are L7 governance; internal use history is L6; interpretation is L5; no local L0 object exists unless this bounded thing authors a semantic version.
Publication	Production is L4; scholarly/research meaning is L5; internal history is L6; provenance, public/legal governance, and the released publication object are L7.

Author note. Writing is L4 as work. An exemplar/model/thesis/file is made here; its research meaning may be L5 and its released publication may be L7.

L5 — RESEARCH / DISCOURSE

Interpretation, review, judgment, argument, and teaching.

Core question	What may responsibly be concluded, questioned, taught, compared, or decided from the available work?	
Function	Frame questions, interpret evidence, evaluate claims, conduct review, govern research decisions, and communicate scientific meaning.	
Feeds	Questions/needs from L7 plus L4 work/evidence, L3/L2 theoretical objects, and L1/L0 dependencies as the inquiry requires.	
Output	A bounded claim/decision with cited dependencies, limitations, review status, and reproducibility information for L6 retention and L7 presentation.	
Hard boundary	Consensus, prose, authority, or usefulness cannot substitute for proof, measurement, or a lower-layer fact.	
Machine metadata	Machine-readable metadata may identify workflow state, ACL/role constraints, review status, routing, institutional access, and release readiness. These instructions govern machine handling; they do not create the research meaning.	

Owns / represents	Permitted actions	Acceptance / boundary check
- research questions, hypotheses, rationale, analyses, interpretations, findings - literature search and relevance judgment - claim scope, limitations, uncertainty, explanation, conclusion - review, audit, research governance, teaching, manuscript meaning	- compare claim to evidence; interpret; criticize; accept/reject/qualify/supersede - request revision from a lower owner without silently mutating it - prepare a claim for preservation and presentation	- Claim is no broader than evidence/maturity. - Methods, exclusions, uncertainty, and inferential limits are disclosed. - Research meaning is distinguished from L4 writing and L7 release.

Directory structure



Boundary / traversal rule. L5 may interpret externally owned definitions, but interpretation does not transfer ownership of the peer definition into L0.

Directory prose and rationale

Directory / thing	Prose	Why
uso/l5/	Layer 5 container for research interpretation, review, judgment, argument, research governance, and teaching.	Keeps scientific meaning and decision distinct from the L4 work product and the L7 presentation surface.
research/	Directory for the active research record and discourse owned at L5.	Keeps inquiry and interpretation separate from institutional participant structure.
research/lanes/	Directory for distinct lines of inquiry within the research record.	Prevents separate questions or investigative paths from being collapsed into one undifferentiated research stream.
research/review/	Directory for review, criticism, audit, and qualification of research claims.	Keeps review activity distinguishable from the findings being reviewed.
research/findings/	Directory for findings that may responsibly be stated from the available work and evidence.	Keeps stated findings tied to L5 judgment rather than treating raw L4 output as a conclusion.
institution/	Directory for the institutional research setting and participant positions.	Keeps the social/institutional setting distinct from the research record itself.
institution/peers/	Directory for peer participant positions within the institutional setting.	Keeps peer participation distinct from other institutional roles.
institution/undergraduates/	Directory for undergraduate participant positions within the institutional setting.	Keeps undergraduate participation distinct from peer and research-owner roles.

In the case of:

Forward-facing discourse forums, chat, and institutional documentation may allow fellows or participants to operate under a shared institutional identity while access controls separate participant roles and lower-layer discourse backends.

Standing. A forum or chat interface may be realized operationally at L4 and exposed at L7, while institutional roles, discourse, interpretation, review, and research meaning carried through it remain L5-owned. L7 governance concerns the external/public legal and access relation, not scientific judgment.

Author note. External definitions can be interpreted here. Interpretation does not transfer ownership of the peer definition into L0.

L6 — LIBRARY

Memory, history, reconstruction, retirement, and query orientation.

Core question	Can the object's internal history, prior states, failures, corrections, and retirement be reconstructed and correctly oriented for inspection?	
Function	Preserve internal memory, prior versions, branch history, corrections, failures, supersession, retirement, ledgers, and reconstructable relations without re-owning external provenance.	
Feeds	L7 provenance/governance/boundary events plus versioned relations and artifacts from L5 through L0.	
Output	A reconstructable Library record identifying the relevant prior state, branch, correction, failure, supersession, retirement, and internal relation needed for inspection or reproduction.	
Hard boundary	L6 records, preserves, and orients; it does not make the L5 research judgment or establish truth.	
Machine metadata	Machine-readable metadata may identify retention, archival, retirement, supersession, reconstruction, and integrity instructions for memory/history. References to external provenance resolve to L7 rather than being re-owned by L6.	
Owns / represents	Permitted actions	Acceptance / boundary check
- prior internal states and versions - branch history and decision history - failed paths and unresolved obligations - corrections, supersession, retirement, and retained memory	- record, preserve, compare, branch, supersede, retire, reconstruct - retain prior classifications and corrections without erasure - orient inspection before entering the active payload	- Prior internal states and changes are reconstructable. - Correction does not require deletion of prior state. - External source/provenance references resolve back to L7. - Retirement preserves the record rather than erasing it.

Directory structure

```
uso/l6/0
├─ history/0
└─ retired/0
```

Boundary / traversal rule. For an externally encountered object, L7 is the boundary and owns its external provenance/governance relation. L6 is the first interior layer used to reconstruct prior internal states, changes, failures, supersession, and retirement before deeper inspection.

Directory prose and rationale

Directory / thing	Prose	Why
uso/l6/	Layer 6 Library container for internal memory, history, prior states, failures, corrections, supersession, retirement, and reconstruction.	Keeps reconstructable scientific memory inside the USO while external provenance and public/legal relations remain on the L7 forward face.
history/	Directory for prior states, branch history, corrections, failures, unresolved paths, and reconstructable development over time.	Preserves how the object changed instead of presenting only the latest state.
retired/	Directory for retired or superseded internal states and bindings that are no longer current.	Implements retirement as preservation rather than deletion, including the history of peer bindings that are no longer active.

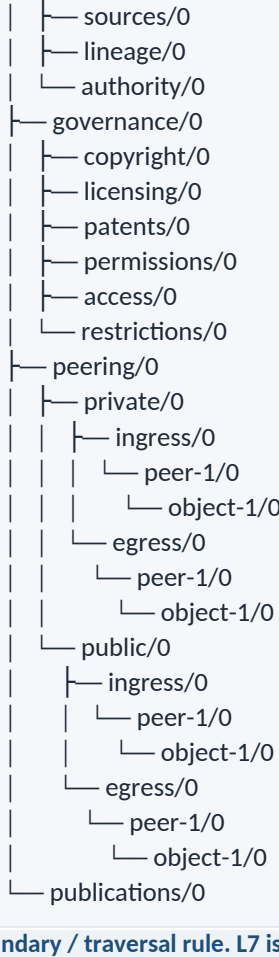
Author note. When full interpretation begins outside the object, L7 is encountered first and L6 is the first interior layer. Interior-only inspection can therefore be described as beginning at L6.

L7 — SURFACE / FORWARD FACE

External provenance, public/legal governance, peering, publications, exposure, and exchange.

Core question	What provenance, governance, peer relation, publication, or other external representation must exist at the object's forward face?	
Function	Provide the external Surface for provenance, prior art, public/legal governance, peering, publications, and release without absorbing external authority or granting lower-layer validity.	
Feeds	Incoming peer objects, source/prior-art/authority information, applicable public/legal governance requirements, and approved L6-backed outward release packages.	
Output	A bounded provenance record, governance record, ingress/egress binding, or publication/presentation with explicit source, authority, permissions, status, and direction where applicable.	
Hard boundary	L7 owns the external relation, not the lower-layer scientific meaning. Presence, publication, copyright, patent status, custody, or provenance at L7 does not make an object valid, complete, true, or internally authored.	
Machine metadata	Machine-readable metadata may identify provenance bindings, public/legal governance, visibility, ingress/egress direction, peer/object identity, release state, permissions, and NO EDIT/NO DELETE constraints for peer-owned ingress. Metadata does not transfer semantic authority.	
Owns / represents	Permitted actions	Acceptance / boundary check
- provenance: prior art, sources, lineage, authority - public/legal governance: copyright, licensing, patents, permissions, access, restrictions - peering: exposure/custody, ingress/egress, peer/object bindings - publications and release identity	- receive, bind, attribute, disclose, present, publish, release, route - record provenance and prior art - apply public/legal permissions and restrictions - authorize, deny, quarantine, retire a binding without deleting its history	- Source, version, authority, and provenance are explicit where required. - Applicable public/legal governance is explicit where required. - External ingress is not silently admitted to L6-L0. - Publication or legal status does not strengthen scientific evidence. - Unused canonical branches may remain empty without implying failure.

Directory structure



Boundary / traversal rule. L7 is the forward face outside the USO. It owns the external relations of the bounded object: provenance, governance, peering, and publication state. Public/private exposure and ingress/egress direction remain independent peering distinctions. L7 does not thereby acquire or change the lower-layer scientific meaning.

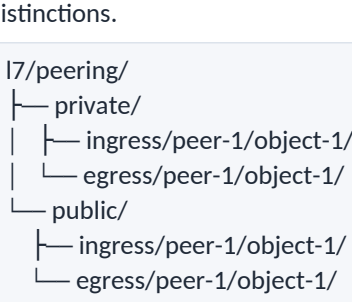
Directory prose and rationale

Directory / thing	Prose	Why
I7/	External Surface / forward-face container outside the USO where the bounded object meets external people, institutions, systems, repositories, legal/public conditions, publications, and data sources.	Keeps external relations separate from the scientific interior while giving those relations an explicit owner.
provenance/	Directory for the external provenance of the bounded object: where it came from, what preceded it, what source/version is being used, and what authority is being represented.	Places externally meaningful source and lineage information at the forward face rather than burying it inside internal memory.
provenance/prior-art/	Directory for identified prior art and antecedent work relevant to the bounded object.	Makes the antecedent landscape explicit without confusing prior art with the object's locally authored L0 meaning.
provenance/sources/	Directory for source identity and exact source/version references.	Keeps source identification inspectable at the external face.
provenance/lineage/	Directory for externally relevant lineage and derivation relationships.	Makes origin and relation traceable without turning lineage into scientific validity.
provenance/authority/	Directory for represented source/owner authority associated with the external object or record.	Distinguishes who owns or speaks for an external object from what the local research concludes about it.
governance/	Directory for public/legal/custodial conditions attached to externally exposed or received objects.	Separates legal and access conditions from provenance, peering direction, publication, and scientific judgment.
governance/copyright/	Directory for copyright-related representation when applicable.	Copyright governs external use of an artifact; it does not change the artifact's scientific meaning.
governance/licensing/	Directory for licensing terms or status when applicable.	Keeps permissions of use explicit at the external face.
governance/patents/	Directory for patent-related representation when applicable.	Provides an agnostic forward-face position for patent status or materials without requiring every project to use it.
governance/permissions/	Directory for explicit permissions attached to external use, disclosure, or exchange.	Keeps permission separate from ownership and from scientific validity.
governance/access/	Directory for access conditions applicable to externally facing objects.	Makes who may access an object a boundary condition rather than an internal scientific claim.
governance/restrictions/	Directory for restrictions that constrain external disclosure, use, transfer, or handling.	Keeps restrictions explicit and machine-addressable without redefining the underlying research.
peering/	Directory for relationships to externally owned peers and the bounded objects exchanged with them.	Preserves external ownership and authority instead of absorbing the peer into the USO.
private/	Peering branch for non-public exposure or custody.	Keeps exposure/custody distinct from direction.
private/ingress/	Private inbound peering position.	Represents something entering from a peer without making it public or internally owned.
private/egress/	Private outbound peering position.	Represents something leaving toward a peer without making it a public release.
public/	Peering branch for public exposure or custody.	Keeps public exposure distinct from whether the object is entering or leaving.
public/ingress/	Public inbound peering position.	Represents a publicly exposed external object entering the local boundary without transferring ownership.
public/egress/	Public outbound peering position.	Represents public outward exchange toward a peer; the released publication object itself is held by the L7 publications branch.
*/peer-1/	Directory identifying the external peer within an ingress or egress position.	Keeps peer identity distinct from the particular object exchanged with that peer.
*/peer-1/object-1/	Directory identifying one bounded peer-owned or peer-directed object.	Allows one peer to expose or receive multiple objects without treating the peer itself as the object. Peer-owned ingress is reference/binding only: NO EDIT and NO DELETE by the receiving repository.
publications/	Directory for released publication objects at the forward face.	Keeps publication release at L7 while writing remains L4, research meaning remains L5, and internal history remains L6.

In the case of:

L7 scientific desktop and forward-face semantics

The Surface is organized around four owned external relations: provenance, governance, peering, and publications. Within peering, exposure/custody (public versus private) and direction (ingress versus egress) remain independent semantic distinctions.



Public ingress is the natural peering position for externally owned, publicly available definitions or peer objects. Private ingress is available for controlled inbound material. Public egress handles outward public exchange toward peers; private egress supports restricted outbound exchange. Released publications are represented by the L7 publications branch, while provenance and public/legal governance remain separately addressable.

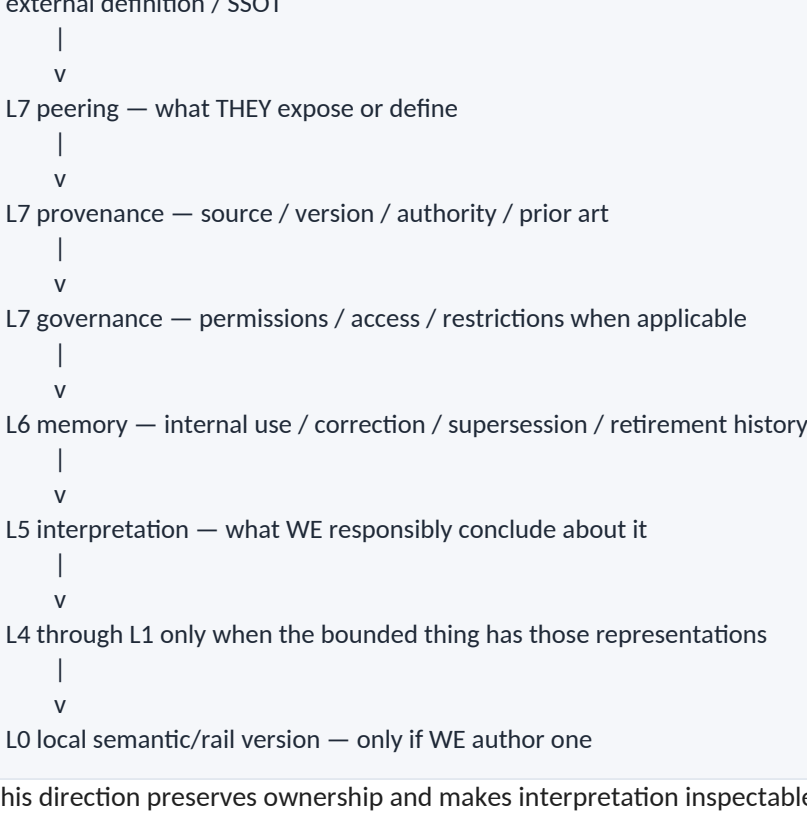
Peer-object contract. peer-1 identifies the external peer and object-1 identifies the bounded object exposed by that peer. The object may be a document, database, foreign/federated data endpoint, or repository. A peer-owned ingress object is bound/reference-only: NO EDIT and NO DELETE by the receiving repository. Retirement or supersession of the internal binding history is preserved at `uso/l6/retired/` rather than erased.

Author note. The agnostic L7 topology carries admissible external-facing capabilities even when a project does not use them. A classroom experiment may leave governance and publication branches empty; a science fair, public release, publication, patent, or controlled exchange can populate the relevant forward-face positions without changing the architecture.

External definitions and inward interpretation

A definition lives once at its owner. When 7LM does not author the definition, it remains bound to the external owner at L7 peering rather than being copied into L0. L7 provenance records source/version/authority and prior-art relation; L7 governance records applicable public/legal/access conditions; L6 preserves the internal history of how the binding was used or retired; L5 compares and interprets it. Lower layers are reached only if the bounded object actually establishes corresponding local representations.

Author note. Public/private and ingress/egress are explicit semantic positions in the canonical research tree. Their presence does not assert that every external implementation must use the same filesystem vocabulary.



This direction preserves ownership and makes interpretation inspectable. It also prevents a copied external definition from masquerading as a locally authored semantic object.

Forward-face model

L7 is the desk: the current external surface where finished work, external bindings, provenance, governance, peering, and publications become addressable without moving their scientific meaning out of the USO.

Conceptual distinctions include consumer <-> product; customer <-> service; request <-> response; need <-> offer; incoming object <-> outgoing object; and private custody <-> public exchange.

Presence on the desk establishes nothing by itself. Provenance, custody, authority, copyright, patent status, permissions, internal consistency, and publication each answer different questions; none of them alone establishes scientific truth.

Standing rules

A definition lives once, at its owner. L7 holds the external binding and provenance relation, never a copied substitute for the owner's definition.

To change an immutable external object, return to its owner. The repository using the object does not edit the peer-owned source.

Material crossing outward may lose fidelity and may never gain scientific maturity merely by being exposed.

No outsider traverses the USO by default. What an outside reader or system needs is represented at the forward face.

Machine-readable metadata may support automation, but the architecture does not assume that automation has been enabled or validated. Current research practice may remain manual until the metadata contract is tested.

Appendix A — Retired compact layer sheets

Each layer above is stated once, by its uniform information sheet. The compact prose sheets that previously restated each layer contract are retained here unedited, one per layer, in the order they appeared. Each layer's Author note remains on that layer's page; the retained sheet carries the Directory anchor line it was written with.

Retired. Compact sheet for L0 — LAYER 0, superseded by the uniform sheet on the L0 page. Retained as written.

L0 — LAYER 0

Core question. What local ontological, semantic, epistemic, and universality distinctions does THIS object establish before an admissible state is declared?

Function. Hold the object's locally authored foundational representation and system-level pedagogy without absorbing externally owned definitions.

Feeds. A bounded thing plus external bindings already identified at L7, oriented through L6, and interpreted where necessary at L5.

Output. A local rail-grounded L0 contract sufficient for L1. A rail may remain sparse or gate-only when no local representation has been authored.

Hard boundary. External SSOT definitions remain at L7 peering. No state transition, operator, proof, numerical method, experiment, or research conclusion is imported here.

Owns / represents. Ontological, Semantic, Epistemic, and Universality rail boundaries for this object. System-level traversal/attachment pedagogy as a directory/function, without making Pedagogy a fifth rail. Local vocabulary or semantic representation actually authored by this object.

Permitted actions. bind local distinctions to their owning rail; state ambiguity or unresolved meaning without filling it by guess; version a changed local distinction rather than silently rewriting prior use.

Acceptance. Every L1 distinction can resolve to a local L0 distinction or an explicitly named unresolved dependency. No external definition is copied inward merely to make L0 look complete. No neighboring rail silently supplies another rail's local meaning.

Directory anchor. `uso/l0/{universality,pedagogy,epistemic,ontology,semantics/definitions}/`

Retired. Compact sheet for L1 — STATE SPACE, superseded by the uniform sheet on the L1 page. Retained as written.

L1 — STATE SPACE

Core question. What states and distinctions are possible or admissible under the established L0 contract?

Function. Freeze the board: dimensions, admissible states, invariants, exclusions, equivalence/identity conditions, and static compatibility.

Feeds. The current L0 contract for the bounded thing.

Output. An L1 state-space contract defining what L2 may represent and operate upon.

Hard boundary. No carrying, transition, selection, learning, calculation, propagation, or transformation. Formal operation begins at L2.

Owns / represents. admissible and excluded states; dimensions, domains, ranges, equivalence classes, identity conditions; static invariants and compatibility constraints.

Permitted actions. admit, exclude, distinguish, bound, freeze; test state membership and compatibility; issue a new version when the possibility landscape changes.

Acceptance. Every admitted state is grounded in L0. Invariants are mutually compatible or the collision is explicit. Changes return to L1 and create a new version.

Directory anchor. `uso/l1/state-space/`

Retired. Compact sheet for L2 — FORMAL MATHEMATICS, superseded by the uniform sheet on the L2 page. Retained as written.

L2 — FINITE / FORMAL MATHEMATICS

Core question. What follows exactly when lawful operations are defined over L1?

Function. Define notation, types, relations, operators, constructors, carriers, exact models, derivations, and proof obligations.

Feeds. The L1 state-space contract.

Output. An exact L2 formal contract supplying L3 with behavior, invariants, formulas, and unresolved proof obligations.

Hard boundary. No engineering tolerance, approximation quality, physical realization, measurement, or empirical truth is claimed here.

Owns / represents. formal symbols and types; relations, functions, operators, gates, constructors, carriers, transition systems; axioms, definitions, lemmas, theorems, derivations, exact formulas.

Permitted actions. formalize, derive, prove, type-check, construct, test closure; declare domain/codomain/partiality; mark conjectural or unproved claims explicitly.

Acceptance. Every symbol binds to L0 meaning and L1 admissibility. Every operation has declared domain/codomain or partiality. Exact results are supported by proof/derivation or marked as unproved.

Directory anchor. `uso/l2/formal-mathematics/{relations,operators,proofs}/`

Retired. Compact sheet for L3 — ENGINEERING MATHEMATICS, superseded by the uniform sheet on the L3 page. Retained as written.

L3 — INFINITE / ENGINEERING MATHEMATICS

Core question. How can the exact L2 relation be made feasible under declared constraints?

Function. Translate formal behavior into implementable numerical/engineering methods with explicit units, precision, approximation, tolerance, uncertainty, complexity, and operating regions.

Feeds. L2 formal contract plus declared environmental, resource, and performance constraints.

Output. An L3 engineering specification telling L4 what can be built/run and within what conditions.

Hard boundary. L3 is not the realized L4 output and does not itself create empirical evidence.

Owns / represents. algorithms and abstract data-structure choices; units, numerical representation, precision, approximation, discretization, tolerance; error/uncertainty budgets, sensitivity, stability, convergence, complexity; simulation, optimization, feasibility arguments.

Permitted actions. approximate, discretize, optimize, estimate, bound error, choose representation; compare candidate methods to L2 invariants; state operating assumptions and valid region.

Acceptance. Error/uncertainty, units, precision, and tolerance are explicit. Required L2 invariants are preserved within declared bounds. Resource/complexity claims are analyzed or labeled estimates.

Directory anchor. `uso/l3/engineering-mathematics/{approximation,uncertainty,tolerances}/`

Retired. Compact sheet for L4 — LAYER 4, superseded by the uniform sheet on the L4 page. Retained as written.

L4 — LAYER 4

Core question. What was actually made, written, run, measured, integrated, stored, transported, converted, or delivered?

Function. Commit resources to produce an identifiable intellectual, digital, organizational, or physical work product or recorded failure.

Feeds. Pinned L3–L0 dependency material, exact versions, obligations, demarcation, authority, and resources.

Output. An audit-ready realized work product/evidence record with traceable inputs, transformations, conditions, tests, results, failures, and rework.

Hard boundary. L4 does not authorize its own L5 interpretation and cannot flatten lower contracts into a convenience summary.

Owns / represents. writing/document production as performed work; experiments as executed, observations, measurements, data collection, tests; code execution, storage, deployment, manufacturing, integration, whole systems; resource commitments and work trace.

Permitted actions. make, write, code, run, measure, test, collect, integrate, convert, store, transport, deploy, manufacture; verify conformance and rework against the governing input contract; record result or failure without converting it into a conclusion.

Acceptance. Governing lower dependencies are pinned/resolvable. A distinguishable output or recorded failure exists. Inputs, conditions, transformations, measurements, and demarcation are traceable. Verification is separated from L5 judgment.

Directory anchor. `uso/l4/work/{writing,experiments,computation}/`

Retired. Compact sheet for L5 — RESEARCH / DISCOURSE, superseded by the uniform sheet on the L5 page. Retained as written.

L5 — RESEARCH / DISCOURSE

Core question. What may responsibly be concluded, questioned, taught, compared, or decided from the available work?

Function. Frame questions, interpret evidence, evaluate claims, conduct review, govern research decisions, and communicate scientific meaning.

Feeds. Questions/needs from L7 plus L4 work/evidence, L3/L2 theoretical objects, and L1/L0 dependencies as the inquiry requires.

Output. A bounded claim/decision with cited dependencies, limitations, review status, and reproducibility information for L6 retention and L7 presentation.

Hard boundary. Consensus, prose, authority, or usefulness cannot substitute for proof, measurement, or a lower-layer fact.

Owns / represents. research questions, hypotheses, rationale, analyses, interpretations, findings; literature search and relevance judgment; claim scope, limitations, uncertainty, explanation, conclusion; review, audit, governance, teaching, manuscript meaning.

Permitted actions. compare claim to evidence; interpret; criticize; accept/reject/qualify/supersede; request revision from a lower owner without silently mutating it; prepare a claim for preservation and presentation.

Acceptance. Claim is no broader than evidence/maturity. Methods, exclusions, uncertainty, and inferential limits are disclosed. Research meaning is distinguished from L4 writing and L7 release.

Directory anchor. `uso/l5/{research/{lanes,review,findings},institution/{peers,undergraduates}}/`

Retired. Compact sheet for L6 — LIBRARY, superseded by the uniform sheet on the L6 page. Retained as written.

L6 — LIBRARY

Core question. Can the object, its lineage, and the decisions around it be reconstructed and correctly oriented for inspection?

Function. Preserve provenance, versions, lineage, ledgers, history, source identity, and reconstructable relations, and orient the incoming query to exact objects.

Feeds. L7 custody/boundary events plus versioned relations and artifacts from L5 through L0.

Output. A reconstructable Library record identifying the exact object/version/source and relations needed for inspection or reproduction.

Hard boundary. L6 records, preserves, and orients; it does not make the L5 research judgment or establish truth.

Owns / represents. source identity, authorship, custody, versions, timestamps, authority; acquisition routes, rights/access, fixity, inventories; derivation/use/realization/interpretation/correction/supersession/presentation relations; prior states, failed paths, unresolved obligations, branch history.

Permitted actions. record, link, attribute, preserve, compare, branch, supersede, reconstruct; retain prior classifications/corrections without erasure; orient inspection before entering the active payload.

Acceptance. Every cited/released artifact resolves to an exact version. Origin, transformation, responsibility, and supersession are reconstructable. Correction does not require deletion of prior state.

Directory anchor. `uso/l6/{provenance,history,retired}/`

Retired. Compact sheet for L7 — SURFACE / FORWARD FACE, superseded by the uniform sheet on the L7 page. Retained as written.

L7 — SURFACE / PEERING

Core question. What may enter, leave, bind to, or represent the object at its external face?

Function. Provide the external Surface and peering relation without absorbing external authority or granting lower-layer validity.

Feeds. Incoming peer objects plus approved L6-backed outward release packages.

Output. A bounded ingress binding or egress presentation/publication with explicit source, status, permissions, and direction.⁴

Hard boundary. Presence at L7 does not make an object valid, complete, current, true, or internally owned.

Owns / represents. surface bindings; custody/exposure state and permissions; ingress/egress relation; presentation/release identity; binding to externally owned definitions and objects.

Permitted actions. receive, bind, disclose, present, publish, release, route, authorize, deny, quarantine; identify exact version/maturity/status being presented; preserve external authority instead of copying it inward.

Acceptance. Source binding, version, authority, direction, and permissions are explicit. External ingress is not silently admitted to L6–L0. Publication does not strengthen evidence merely by being public. Unused canonical branches may remain empty without implying failure.

Directory anchor. `l7/peering/{private/{ingress/peer-1/object-1,egress/peer-1/object-1},public/{ingress/peer-1/object-1,egress/{peer-1/object-1,publications}}}/`